



Guided Mutation Proposal for Protein Engineering



Paper

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1 The problem

Goal of protein engineering: climb a rugged fitness landscape, learning from a Deep Mutational Scan (DMS)

wild-type M S K G E E L G V V L D G D V N G H K F S G fitness 1.00

From variants scored experimentally in a DMS...

M S K G E E L G V **A** L D G D V N **V** H K F S G 1.38
M S K G E E L G V V L D G **C** V N G H K F S G 1.46
M S K **A** E E L G V V **I** **I** G D V N G H K F S G 0.77
M S K G E E L G V V L D G D V N G H K F **P** G 0.42

...how can we computationally learn to propose mutations that climb in fitness?

wild-type M S K G E E L G V V L D G D V N G H K F S G 1.00
+ mut 1 M S K G E **I** L G V V L D G D V N G H K F S G 1.62
+ mut 2 M S K G E **I** L G V V L **R** G D V N G H K F S G 2.48
+ mut 3 M S K G E **I** L G V V L **R** G D V N G H K **Y** S G 3.95

Current approaches leverage a single model, creating a tension: learn representation *and* proposal.

Representation (oracle)

- Scores mutation effects
- Wants the evolutionary **average** distribution — what's typical across a protein family

average the norm

Proposal (search)

- Picks which mutations to try
- Wants the **extraordinary** — rare, high-reward variants far from average

chase the rare

single foundation
model (e.g., ESM)

architectural mismatch

That single model is often a masked language model, which must store co-evolution patterns in its weights. In contrast, multiple sequence alignment (MSA) models **observe co-evolution at inference time.**

Single-sequence MLM

sees one sequence with positions masked

interaction?
M ? T L ? A

stores co-evolution in parameters
no homologs to compare against

coupling inferred indirectly

MSA-based model

sees many aligned homologs at once

hom 1 M **R** T L **E** A
hom 2 M **K** S L **D** A
hom 3 L **R** T I **E** A
hom 4 M **K** T L **D** A
K & D and R & E co-vary

coupling read directly

2 AURORA's contributions

MSA-based representations capture epistasis better

MSA Pairformer outperforms ESM on zero-shot and supervised variant prediction, both on the ProteinGym benchmark and on a novel synthetic benchmark

Decoupled proposal via RL enables multi-site mutation

Mutation proposal is an MDP over a fixed oracle; a policy proposes position and amino acid edits jointly. Different oracles and sequence representations yield different appetites for exploration and novelty

Wet-lab validation showcases AURORA's success

Trained on 50,000 variants, AURORA generates novel avGFP variants attaining 3–4× wild-type fluorescence

3 Experiments

Synthetic epistasis benchmark for PLMs

FITNESS OF SEQUENCE $\mathbf{x}$ GIVEN ALIGNMENT $\mathbf{M}$

$$f(\mathbf{x}, \mathbf{M}) = f_{\text{add}}(\mathbf{x}) + f_{\text{epi}}(\mathbf{x}) + f_{\text{cov}}(\mathbf{x}, \mathbf{M}) + f_{\text{int}}(\mathbf{x}, \mathbf{M}) + \epsilon$$

Additive · single-site effects

$$f_{\text{add}}(\mathbf{x}) = \sum_{i=1}^L h_i(x_i)$$

Each position contributes independently via a per-site function h_i

Epistatic · pairwise couplings

$$f_{\text{epi}}(\mathbf{x}) = \sum_{(i,j) \in \mathcal{E}} J_{ij}(x_i, x_j)$$

Interacting position pairs (set $\mathcal{E}$) are coupled by a function $J_{i,j}$

Covariation · alignment signal

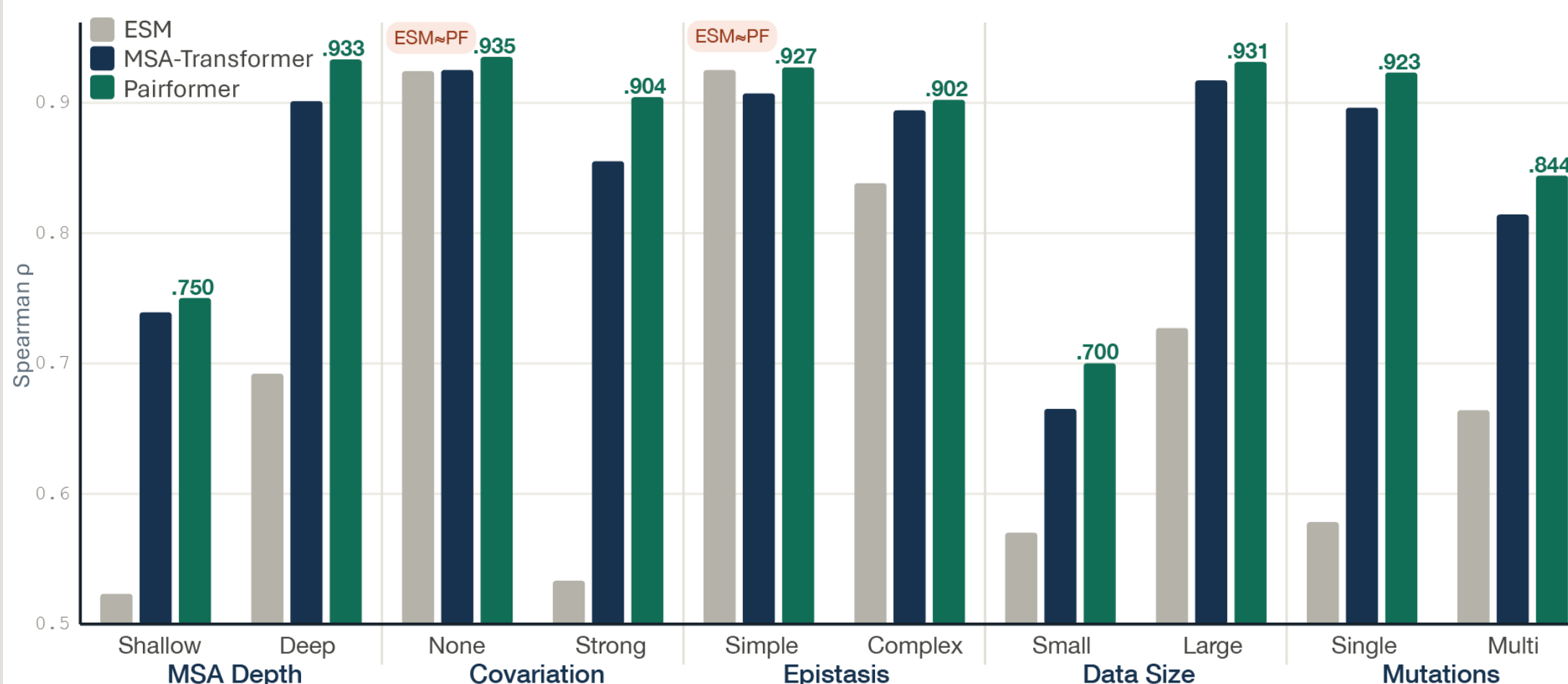
$$f_{\text{cov}}(\mathbf{x}, \mathbf{M}) = \sum_{(i,j) \in \mathcal{E}} w_{ij} C_{ij}(\mathbf{M})$$

Weights $w_{i,j}$ scale the covariation strength $C_{i,j}(\mathbf{M})$ read from $\mathbf{M}$

Interaction · identity × covariation

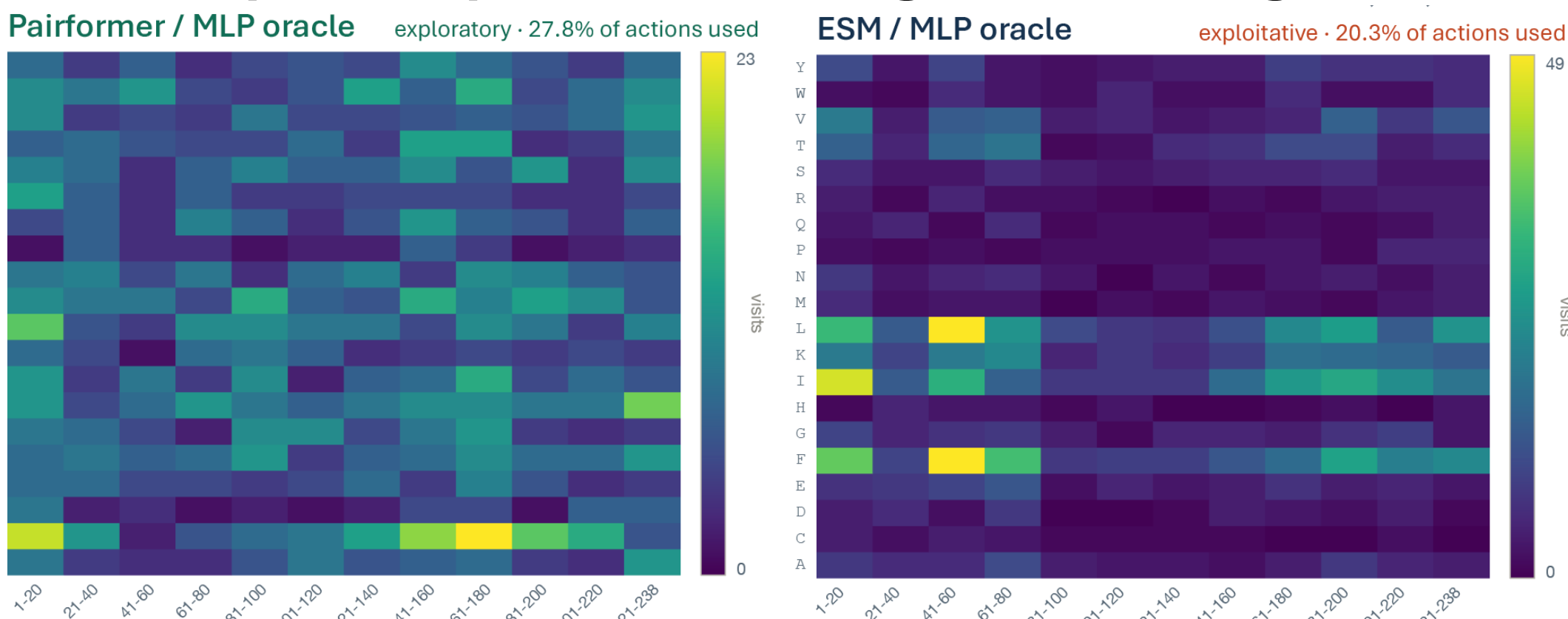
$$f_{\text{int}}(\mathbf{x}, \mathbf{M}) = \sum_{(i,j) \in \mathcal{E}} V_{ij}(x_i, x_j) C_{ij}(\mathbf{M})$$

Couples sequence identity $V_{i,j}$ with evolutionary covariation $C_{i,j}(\mathbf{M})$



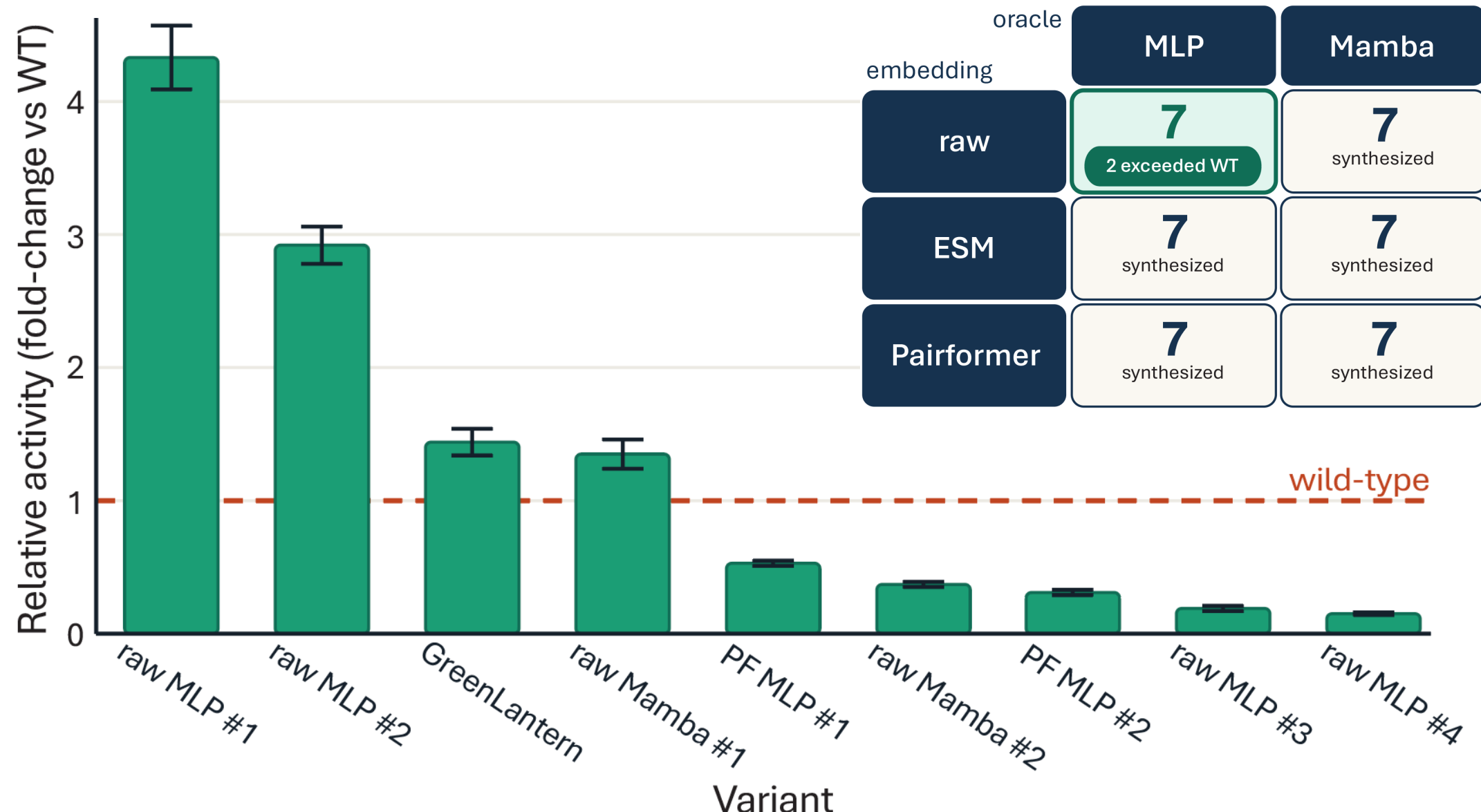
MSA advantage emerges when covariation is strong or signal is sparse, regimes that ESM cannot represent

Action space exploration during PPO training



ESM-guided policy concentrates visits on a few actions; Pairformer-guided policy explores more of action space

Wet lab validation of avGFP mutants



AURORA designs 4x brighter avGFPs; unexpectedly, the policy guided by MLP oracle on raw sequences is best